MIL-W-5086/3B
23 November 1983
SUPERSEDING
MIL-W-5086/3A
2 December 1974
See also "Supersession Data" on Page 3

MILITARY SPECIFICATION SHEET

(B)

WIRE, ELECTRICAL, POLYVINYL CHLORIDE INSULATED, PVC-GLASS-PVC-NYLON, TIN-COATED COPPER CONDUCTOR, 600-VOLT, 105°C

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the wire described herein shall consist of this document and the latest issue of Specification MIL-W-5086.

(B) FROM DATE OF ISSUE OF THIS REVISION, WIRE OF THIS SPECIFICATION SHEET SHALL NOT BE USED IN AEROSPACE APPLICATIONS. SEE "NON-USE" NOTE ON PAGE 3.

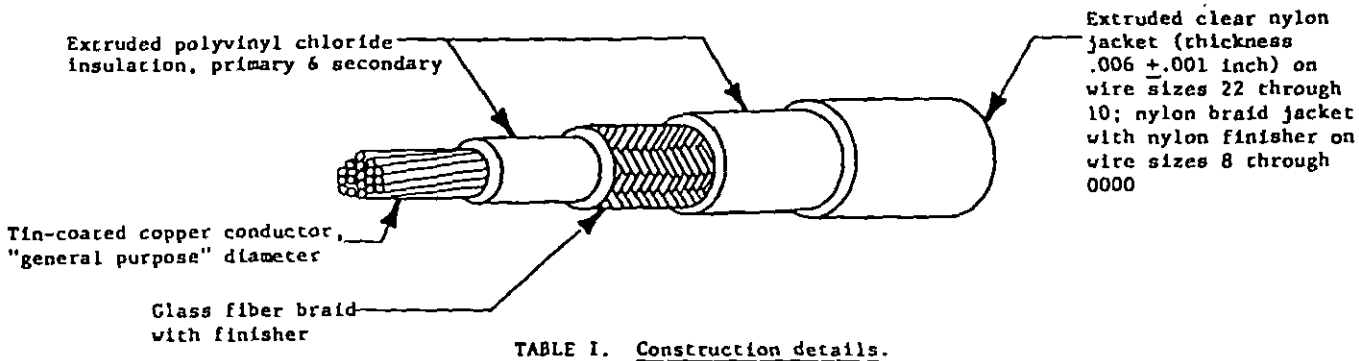


TABLE I. Construction details.

Part no. 1/	Wire size	Stranding (Number of strands X AWG gage of strands)	Diameter of stranded conductor (inches)		Finished wire		
			(min)	(max)	Resistance at 20°C (68°F) (ohms/1000 ft) (max)	Diameter (inches)	Weight (lbs/1000 ft) (max)
M5086/3-22-*	22	19 x 34	.029	.033	16.2	.085 ±.005	5.7
M5086/3-20-*	20	19 x 32	.037	.041	9.88	.095 ±.005	7.6
M5086/3-18-*	18	19 x 30	.046	.051	6.23	.110 ±.005	11.0
M5086/3-16-*	16	19 x 29	.052	.058	4.81	.125 ±.005	14.0
M5086/3-14-*	14	19 x 27	.065	.073	3.06	.143 ±.007	20.0
M5086/3-12-*	12	37 x 28	.084	.090	2.02	.163 ±.007	28.3
M5086/3-10-*	10	37 x 26	.106	.114	1.26	.193 ±.007	44.0
M5086/3-8-*	8	133 x 29	.158	.173	.701	.248 ±.007	70.0
M5086/3-6-*	6	133 x 27	.198	.217	.445	.303 ±.007	113.0
M5086/3-4-*	4	133 x 25	.250	.274	.280	.360 ±.010	170.0
M5086/3-2-*	2	665 x 30	.320	.340	.183	.435 ±.010	250.0
M5086/3-1-*	1	817 x 30	.360	.380	.149	.485 ±.010	310.0
M5086/3-01-*	0	1045 x 30	(B) .395	.425	.116	.535 ±.015	400.0
M5086/3-02-*	00	1330 x 30	(B) .440	.475	.091	.595 ±.015	500.0
M5086/3-03-*	000	1665 x 30	(B) .500	.540	.071	.665 ±.015	620.0
M5086/3-04-*	0000	2109 x 30	(B) .565	.605	.056	.735 ±.015	785.0

1/ Part no.: The asterisks in the part number column, Tables I through III, shall be replaced by color code designators in accordance with MIL-STD-681. Examples: Size 20, white - M5086/3-20-9; white with orange stripe - M5086/3-20-93.

(B) denotes changes

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:16Z
Check the source to verify that this is the current version before use.

(B)

TABLE II. Bend test mandrels and test loads.

Part no.	Mandrel diameter (inches) (±3%)			Test load (lbs) (±3%)	
	Life cycle (oven and bend tests) 1/	Cold bend test	Wrap test	Life cycle (oven and bend tests) 1/	Cold bend test
M5086/3-22-*	4.5	3.0	0.51	.75	2.0
M5086/3-20-*	4.5	3.0	.57	.75	2.0
M5086/3-18-*	4.5	3.0	.66	1.0	2.0
M5086/3-16-*	6.5	3.0	.75	1.0	3.0
M5086/3-14-*	6.5	6.0	.86	1.0	3.0
M5086/3-12-*	6.5	6.0	.98	3.0	3.0
M5086/3-10-*	10.0	6.0	1.16	3.0	5.0
M5086/3-8-*	10.0	6.0	--	3.0	5.0
M5086/3-6-*	10.0	10.0	--	6.0	10.0
M5086/3-4-*	10.0	10.0	--	6.0	10.0
M5086/3-2-*	10.0	18.0	--	6.0	15.0
M5086/3-1-*	10.0	18.0	--	6.0	15.0
M5086/3-01-*	10.0	18.0	--	10.0	20.0
M5086/3-02-*	10.0	18.0	--	10.0	25.0
M5086/3-03-*	10.0	18.0	--	10.0	30.0
M5086/3-04-*	10.0	18.0	--	10.0	30.0

1/ Also for bend tests after immersion.

Wire ratings and additional requirements

Temperature rating: 105°C (221°F) max conductor temperature
Voltage rating: 600 volts (rms) at sea level
Blocking: 105° ±2°C (221° ±3.6°F)
Color: In accordance with MIL-STD-104, Class 1; white preferred
Color striping or banding durability:
    Sizes 22 through 10: 250 cycles (500 strokes) (min), 500 grams weight
    Sizes 8 through 0000: 50 cycles (100 strokes) (min), 250 grams weight
Flammability (Method 1):
    30 sec (max) after-flame
    1.50 inches (max) flame travel, either direction on wire
    No flaming of tissue paper
Humidity resistance:
    Sizes 22 through 10: 100 megohms for 1000 ft, min insulation resistance after humidity exposure
    Sizes 8 through 0000: Not applicable
Identification durability:
    Sizes 22 through 10: 125 cycles (250 strokes) (min), 500 grams weight
    Sizes 8 through 0000: 50 cycles (100 strokes) (min), 250 grams weight
Impulse dielectric test:
    Primary insulation (when test is used in lieu of spark test): 6.0 kilovolts (peak), 100% test
    Finished wire: 8.0 kilovolts (peak), 100% test
Insulation resistance:
    Sizes 22 through 10: 500 megohms for 1000 ft (min)
    Sizes 8 through 0000: Not applicable
Life cycle: Oven temperature, 120° ±2°C (248° ±3.6°F)
Low temperature (cold bend): -55° ±2°C (-67° ±3.6°F)
Shrinkage: 0.125 inch max at 150° ±2°C (302° ±3.6°F)
Smoke: 110°C (230°F)
Spark test of primary insulation:
    Sizes 22 through 10: 3000 volts (rms), 60 Hz, 100% test
    Sizes 8 through 0000: 4000 volts (rms), 60 Hz, 100% test
Surface resistance:
    Sizes 22 through 10: 5 megohm-inches (min), initial and final readings
    Sizes 8 through 0000: Not applicable

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:16Z
Check the source to verify that this is the current version before use.

Thermal shock: Oven temperature, 105° ±2°C (221° ±3.6°F)
Max change in measurement, 0.06 inch
Wet dielectric test: 2000 volts (rms)
Wrap test oven temperature:
Sizes 22 through 10: 95° ±2°C (203° ±3.6°F)
Sizes 8 through 0000: Not applicable

(B) Metric conversion note: Data in this specification sheet may be converted to metric as follows:

Linear dimensions	25.40 x inches = millimeters (mm)
Weight (general)	.4536 x lbs = kilograms (kg)
Wire weight	1.488 x (lbs/1000 ft) = kg/km
Conductor resistance	3.281 x (ohms/1000 ft) = ohms/km
Insulation resistance	.3048 x (megohms for 1000 ft) = megohms for 1 km
Surface resistance	Metric documents may differ from MIL-STD-228 in spacing of the electrodes for this determination and in the method of expressing the test results. Where the electrode spacing is 25.0 millimeters in the metric document, the MIL-W-5086 megohm-inches resistance (defined as total megohms resistance times inches wire diameter) may be converted as follows: 25.0 x (megohm-inches diameter) = megohm-mm diameter 78.5 x (megohm-inches diameter) = megohm-mm circumference 3.14 x (megohm-inches diameter) = megohm-mm circumference per mm of electrode spacing

SUPERSESSION DATA: The wire of this specification sheet replaces and supersedes Type III wire of MIL-W-5086A (superseded 19 March 1968) and MS25190 (canceled 29 May 1969). Supersession by part number is in accordance with Table III.

TABLE III. Supersession by part number.

Size designation, MIL-W-5086A Type III	Part number, MS25190 Type III	Part number, MIL-W-5086/3
AN-22	MS25190-C-22	M5086/3-22-*
AN-20	MS25190-C-20	M5086/3-20-*
AN-18	MS25190-C-18	M5086/3-18-*
AN-16	MS25190-C-16	M5086/3-16-*
AN-14	MS25190-C-14	M5086/3-14-*
AN-12	MS25190-C-12	M5086/3-12-*
AN-10	MS25190-C-10	M5086/3-10-*
AN-8	MS25190-C-8	M5086/3-8-*
AN-6	MS25190-C-6	M5086/3-6-*
AN-4	MS25190-C-4	M5086/3-4-*
AN-2	MS25190-C-2	M5086/3-2-*
AN-1	MS25190-C-1	M5086/3-1-*
AN-0	MS25190-C-01	M5086/3-01-*
AN-00	MS25190-C-02	M5086/3-02-*
AN-000	MS25190-C-03	M5086/3-03-*
AN-0000	MS25190-C-04	M5086/3-04-*

(B) NON-USE AND REPLACEMENT OF MIL-W-5086/3 WIRE IN AEROSPACE APPLICATIONS: The exclusion of MIL-W-5086/3 wire from aerospace applications (page 1 of this document) is in consonance with the provisions of MIL-STD-454 and several military aerospace directives concerning polyvinyl chloride materials.

Replacement wires for the MIL-W-5086/3 items for aerospace applications should be selected from the lists of approved wires in the latest issue of MIL-W-5088, Wiring, Aerospace Vehicle, with due regard to the weight, dimensional, and functional requirements of the specific project or application.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:16Z
Check the source to verify that this is the current version before use.

INTERNATIONAL STANDARDIZATION: Certain provisions of Table I of this specification sheet are the subject of international standardization agreement (ASCC Air Std 12/13). When amendment, revision, or cancellation of this specification sheet is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodians:
Navy - AS
Army - CR
Air Force - 85

Review activities:
Navy - OS
Army - AR, ME, MI
Air Force - 99
DLA - IS
NSA

User activities:
Navy - EC, MC, SH
Army - AT, AV
Air Force - 11

Preparing activity:
Navy - AS
(Project No. 6145-0822-3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:16Z
Check the source to verify that this is the current version before use.